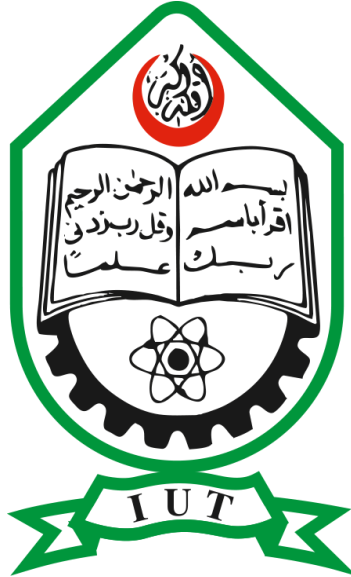




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Artificial Intelligence and Machine Learning Lab

Open-Ended Machine Learning Project

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1. Problem Definition

Real-World Problem

Cardiac arrhythmia—defined as irregular or abnormal heart rhythms—is one of the leading causes of sudden cardiac death worldwide. Conditions such as atrial fibrillation, ventricular tachycardia, and other rhythm disorders can rapidly become life-threatening if not detected early. According to global health statistics from organizations like the World Health Organization, cardiovascular diseases account for millions of deaths each year, with arrhythmias playing a significant role.

Electrocardiography (ECG) is the primary diagnostic tool used to monitor and analyze heart activity. However, accurate interpretation of ECG signals requires trained cardiologists and involves analyzing thousands of heartbeats per patient. This process is not only time-consuming but also prone to human error due to fatigue, variability in expertise, and subjective judgment.

In resource-constrained settings such as Bangladesh, the challenge is even more critical. There is a shortage of specialized cardiologists, especially in rural and remote areas, which limits timely diagnosis and intervention. As a result, many patients remain undiagnosed until severe complications arise.

Why It Matters and How Machine Learning Helps

ECG is a low-cost, non-invasive, and widely available diagnostic method, making it ideal for large-scale screening. However, the sheer volume and complexity of ECG data make manual analysis inefficient. This is where machine learning (ML), particularly deep learning, offers a transformative solution.

Modern ML models—such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)—can automatically learn complex patterns and features directly from raw ECG waveforms. These models are capable of detecting subtle morphological variations (e.g., P-wave abnormalities, QRS complex distortions, irregular RR intervals) that may be difficult for the human eye to consistently identify. Additionally, ML systems provide:

- **Scalability:** Ability to process large volumes of ECG data quickly
- **Consistency:** Eliminates variability caused by human fatigue or subjectivity
- **Accessibility:** Enables deployment in remote or underserved regions via portable devices
- **Real-time analysis:** Supports immediate clinical decision-making and early intervention

By integrating ML into healthcare systems, we can build intelligent diagnostic tools that assist clinicians rather than replace them—enhancing both speed and accuracy of diagnosis.

Project Objectives and Tasks

This project focuses on developing a robust, high-accuracy ECG classification system using benchmark datasets to address two key real-world tasks:

Task 1: Multi-Class Arrhythmia Classification (MIT-BIH Dataset)

Using the MIT-BIH Arrhythmia Database, the system classifies heartbeats into five categories:

- Normal (N)
- Supraventricular (S)
- Ventricular (V)
- Fusion (F)
- Unknown (Q)

This task aims to replicate real clinical scenarios where multiple types of arrhythmias must be distinguished accurately.

Task 2: Binary Classification (PTB Dataset)

Using the PTB Diagnostic ECG Database, the system performs:

- Normal vs. Abnormal classification

This serves as a screening-level diagnostic tool to quickly identify whether further medical evaluation is required.

2. Dataset Preparation

Data Sources

This project utilizes two well-established and publicly available ECG benchmark datasets obtained from Kaggle:

- **MIT-BIH Arrhythmia Database**
 - `mitbih_train.csv` → 87,554 samples
 - `mitbih_test.csv` → 21,892 samples
- **PTB Diagnostic ECG Database**
 - `ptbdb_normal.csv` → 4,046 samples
 - `ptbdb_abnormal.csv` → 10,506 samples

These datasets are widely used in biomedical signal processing research and provide a reliable foundation for training and evaluating machine learning models for arrhythmia detection.

Dataset Characteristics

Each dataset is structured such that:

- Each row represents a single heartbeat signal
- Each sample contains 187 time-step values, corresponding to ECG signal amplitudes over time
- The final column represents the class label

MIT-BIH Dataset

- Multi-class classification problem with 5 distinct classes:
 - Normal (N)
 - Supraventricular (S)
 - Ventricular (V)
 - Fusion (F)
 - Unknown (Q)
- The dataset is highly imbalanced, with the majority of samples belonging to the Normal class, which can bias model predictions if not handled properly.

PTB Dataset

- Binary classification problem:
 - Normal
 - Abnormal
- Compared to MIT-BIH, this dataset is relatively simpler but still exhibits class imbalance, with abnormal samples outnumbering normal ones.

Data Preprocessing Pipeline

To ensure robust and unbiased model performance, several preprocessing steps were carefully implemented:

1. Data Loading

- All CSV files were loaded using no header configuration, as the datasets do not include column names.
- Each dataset was verified for correct dimensionality and structure.

2. Missing Value Analysis

- A thorough inspection of all datasets revealed:
 - No missing or null values in any of the samples
- This eliminated the need for imputation or data cleaning procedures.

3. Handling Class Imbalance

- Both datasets, particularly MIT-BIH, exhibit significant class imbalance, which can lead to poor generalization for minority classes.
- To address this:
 - **SMOTE (Synthetic Minority Over-sampling Technique)** was applied
 - Implemented using **imblearn.pipeline** within each **cross-validation fold**
- This approach ensures:
 - Balanced class distribution during training
 - No data leakage, as oversampling is performed only on training folds, not on validation/test data
- Importantly:
 - The original training set distribution was preserved for final evaluation on the test set to maintain realistic performance metrics.

4. Feature Scaling

- Feature scaling was applied selectively depending on the model:
 - **StandardScaler** used for models sensitive to feature magnitude (e.g., Logistic Regression, SVM, Neural Networks)
 - For models like **Random Forest**, scaling is not strictly required but was applied where beneficial for consistency
- Scaling ensures:
 - Faster convergence during training
 - Improved numerical stability
 - Better model performance in gradient-based algorithms

5. Train-Test Strategy

- The datasets come with predefined official splits, which were used directly:
 - No additional random splitting was performed
- This ensures:
 - Fair comparison with existing research
 - Reproducibility of results

6. Label Encoding

- For deep learning models:
 - Class labels were converted using one-hot encoding via `to_categorical`
- This is essential for:
 - Multi-class classification tasks (MIT-BIH)
 - Proper loss computation using categorical cross-entropy
- For traditional ML models:
 - Labels were kept in integer-encoded format

3. Model Development

To effectively address both multi-class and binary ECG classification tasks, a combination of classical machine learning and deep learning models was implemented. The selection of models was guided by performance, interpretability, computational efficiency, and suitability for time-series ECG data.

3.1 Algorithms Selected

Task 1: Multi-Class Classification (MIT-BIH Dataset)

Using the MIT-BIH Arrhythmia Database, three different models were developed and compared:

Baseline Model: Random Forest

- A **Random Forest classifier** was used as a baseline to establish a performance benchmark.
- Configuration:
 - Number of trees: 100 estimators
 - Maximum depth: 20
 - Class weighting: 'balanced' (to address class imbalance)
- Advantages:
 - Robust to noise and overfitting
 - Handles high-dimensional data well
 - Provides feature importance insights
- This model serves as a reference point to evaluate the effectiveness of deep learning approaches.

Deep Learning Model 1: 1D Convolutional Neural Network (CNN)

- A **1D CNN architecture** was designed to automatically extract spatial features from ECG signals.
- Architecture components:
 - Multiple **Conv1D layers** for feature extraction
 - **MaxPooling layers** to reduce dimensionality
 - **Dropout layers** to prevent overfitting
 - Fully connected (**Dense layers**) for classification
- Model size:
 - Approximately 880,000 trainable parameters
- Strengths:
 - Captures local patterns in ECG signals (e.g., QRS complex, P-wave morphology)
 - Efficient for time-series data
 - Learns hierarchical feature representations automatically

Deep Learning Model 2: Hybrid CNN + LSTM

- A hybrid model combining:
 - **Conv1D layers** (for spatial feature extraction)
 - **LSTM (Long Short-Term Memory)** layers (for temporal dependencies)
- Architecture flow:
 - Conv1D → Feature maps → LSTM → Dense → Output
- Model size:
 - Approximately 79,000 parameters (significantly lighter than standalone CNN)
- Advantages:
 - Captures both:
 - **Local morphological features** (via CNN)
 - **Sequential dependencies over time** (via LSTM)
 - More efficient and less prone to overfitting due to reduced parameter count

Task 2: Binary Classification (PTB Dataset)

Using the PTB Diagnostic ECG Database:

Lightweight 1D CNN

- A simplified CNN architecture was used due to:
 - Smaller dataset size
 - Binary classification nature
- Key features:
 - Fewer convolutional layers
 - Reduced number of parameters
 - Faster training and inference
- Goal:
 - Achieve high accuracy while maintaining computational efficiency, making it suitable for real-time or embedded applications.

3.2 Training Strategy and Techniques

To ensure fair evaluation, prevent overfitting, and improve generalization, several training strategies were applied:

1. Train-Test Strategy

- Used official dataset splits (80 for Training / 20 for testing) provided in the datasets
- No additional random splitting was performed
- Ensures:
 - Reproducibility
 - Fair comparison with existing studies

2. Cross-Validation (Random Forest)

- Applied **5-fold Stratified K-Fold Cross-Validation** for the Random Forest model:
 - Maintains class distribution in each fold
 - Provides more reliable performance estimates
- Integrated **SMOTE** within the pipeline:
 - Applied only to training folds
 - Prevents data leakage
 - Ensures balanced learning

3. Optimization and Loss Functions

- Optimizer:
 - **Adam optimizer** (adaptive learning rate, efficient convergence)
- Loss functions:
 - **Categorical Cross-Entropy** was used for multi-class classification (MIT-BIH)
 - **Binary Cross-Entropy** was used for PTB dataset

4. Regularization and Callbacks

To enhance model performance and prevent overfitting:

- **EarlyStopping**
 - Monitors validation loss
 - Stops training when performance stops improving
 - Prevents unnecessary training and overfitting
- **ReduceLROnPlateau**
 - Dynamically reduces learning rate when validation performance plateaus
 - Helps models converge to better minima
- Important implementation detail:
 - Fresh callback instances were used for each model
 - Prevents unintended state leakage between training runs

5. Reproducibility

To ensure consistent and reproducible results:

- Fixed random seeds:

```
np.random.seed(42)
tf.random.set_seed(42)
```

- Benefits:
 - Deterministic behavior
 - Consistent model initialization and training results

3.3 Implementation Environment

- All models were implemented using:
 - TensorFlow / Keras API
 - Developed and executed in Jupyter Notebook
- Advantages of this setup:
 - Modular experimentation
 - Easy visualization of results
 - Efficient debugging and iteration

4. Performance Evaluation

To comprehensively evaluate model performance, multiple standard classification metrics were used. These metrics provide a balanced assessment, particularly important for imbalanced datasets such as the MIT-BIH Arrhythmia Database.

4.1 Evaluation Metrics

The following metrics were considered:

- **Accuracy**
Measures overall correctness but may be biased toward majority classes.
- **Macro Precision**
Average precision across all classes, treating each class equally.
- **Macro Recall**
Measures the model's ability to correctly identify all classes, especially minority ones.
- **Macro F1-score**
Harmonic mean of precision and recall; a key metric for imbalanced classification.
- **ROC-AUC (macro-averaged)**
Evaluates class separability across all classes.

For the binary classification task using the PTB Diagnostic ECG Database, additional metrics were included:

- **Precision-Recall AUC (PR-AUC)**
More informative for imbalanced binary classification problems.

4.2 Task 1: MIT-BIH Multi-Class Results (Test Set)

Model	Accuracy	Macro Precision	Macro Recall	Macro F1-score	ROC-AUC
Random Forest (Baseline)	0.9757	0.8584	0.8943	0.8715	0.9910
1D CNN	0.9848	0.9027	0.9302	0.9157	0.9941
1D CNN + LSTM	0.9767	0.8548	0.9297	0.8881	0.9935

4.3 Cross-Validation Results (Random Forest)

- **5-Fold Stratified Cross-Validation (with SMOTE inside each fold):**
 - **Macro F1-score = 0.8773 ± 0.0044**

Interpretation

- Very low variance indicates:
 - **High stability**
 - **Strong generalization capability**
- Confirms that the baseline model is reliable and not overfitting

4.4 Task 2: PTB Binary Classification Results (Test Set)

- **Accuracy:** 0.9894
- **Precision:** 0.9924
- **Recall:** 0.9929
- **ROC-AUC:** 0.9978
- **PR-AUC:** 0.9988

Interpretation

- Near-perfect scores indicate:
 - Excellent discrimination between normal and abnormal ECG signals
 - Strong suitability for real-world screening systems

4.5 Class-Level Performance Analysis

Detailed classification reports reveal important insights:

Random Forest

- Normal class: Precision ≈ 0.99 , Recall ≈ 0.99
- Fusion class (weakest): Precision = 0.56, Recall = 0.81

1D CNN (Best Model)

- Normal class: Precision ≈ 0.99 , Recall ≈ 0.99
- Fusion class: Precision = 0.75, Recall = 0.85 (**significant improvement**)

1D CNN + LSTM

- Normal class: Precision ≈ 0.99 , Recall ≈ 0.98
- Fusion class: Precision = 0.68, Recall = 0.85

5. Discussion

5.1 Interpretation of Results

The results clearly indicate that the **1D CNN model** is the most effective for ECG-based arrhythmia classification. On the MIT-BIH Arrhythmia Database, it achieved:

- 98.48% Accuracy
- 0.9027 Macro Precision
- 0.9302 Macro Recall
- 0.9157 Macro F1-score
- 0.9941 ROC-AUC

This demonstrates not only high overall accuracy but also a strong balance between precision and recall across all classes.

The CNN + LSTM hybrid model, while conceptually powerful, did not outperform the standalone CNN. This suggests that:

- Single-beat classification benefits more from spatial feature extraction (CNN)
- Temporal modeling (LSTM) becomes more useful in multi-beat or sequence-based tasks

The Random Forest model provided a strong and stable baseline. Its cross-validation results (0.8773 ± 0.0044 Macro F1) closely align with test performance, confirming:

- Good generalization
- No significant overfitting

For the PTB Diagnostic ECG Database, the lightweight CNN achieved near-perfect binary classification, making it highly effective for:

- Initial screening systems
- Real-time diagnostic support tools

5.2 Class-Level Insights

- **Normal Class**
 - All models performed exceptionally well (Precision & Recall ≈ 0.99)
 - Indicates strong ability to recognize standard ECG patterns
- **Fusion Class (Most Challenging)**
 - Represents mixed morphology (Normal + Ventricular)
 - Most difficult to classify
 - 1D CNN significantly improved performance (Precision = 0.75)
- **Supraventricular Class**
 - Moderately difficult
 - 1D CNN achieved the best recall (~ 0.86), indicating better detection capability

5.3 Strengths of the Approach

- **High performance across all metrics**, especially with the 1D CNN
- **Balanced classification** demonstrated by strong Macro F1-score
- **Proper handling of class imbalance** using SMOTE within cross-validation folds (no data leakage)
- **Excellent ROC-AUC values (>0.99)** indicating strong separability
- **Stable cross-validation results**, confirming robustness
- **Lightweight architectures**, suitable for real-time ECG monitoring systems

5.4 Limitations

- **Lower precision for difficult classes**
 - Fusion and Supraventricular classes remain challenging
- **Lack of temporal context**
 - Models operate on single beats
 - Do not consider RR intervals or rhythm patterns (important for conditions like atrial fibrillation)
- **Dataset constraints**
 - Evaluated on clean benchmark datasets
 - Real-world ECG signals include:
 - Noise
 - Baseline drift
 - Motion artifacts
- **Hybrid model limitations**
 - CNN + LSTM showed slightly lower precision on minority classes
 - Indicates need for better tuning or sequence-based input

5.5 Future Improvements

- **Incorporate temporal information**
 - Use sequences of 5–10 consecutive beats
 - Apply LSTM or Transformer architectures
- **Advanced imbalance handling**
 - Use **Focal Loss** or class-weighted loss
 - Combine with SMOTE for better minority class learning
- **Real-world validation**
 - Test on noisy ECG data
 - Evaluate on datasets from hospitals in Bangladesh
- **Model ensembling**
 - Combine predictions from Random Forest and CNN
 - Improve robustness and generalization
- **Practical deployment**
 - Develop applications using Streamlit or Flask
 - Enable doctors to upload ECG signals and receive instant predictions

5.6 Final Insight

Overall, the results confirm that deep learning—particularly 1D CNNs—offers a powerful, reliable, and scalable solution for ECG-based arrhythmia detection, with strong potential for real-world clinical deployment when combined with further improvements in robustness and temporal modeling.